CS 532, Spring 2019: Quiz 2

NAME:

Problem 1. Consider a digital camera with pixel resolution of 640 pix. wide by 480 pix. tall and a horizontal field of view of 90 degrees. What is the camera's focal length (expressed in pixels)? Make a sketch.

Problem 2. Write the calibration matrix $\mathbf{K}$ for the camera.

Problem 3. In the expression $\mathbf{x} = \mathbf{K}\,[\mathbf{R}|\mathbf{t}]\,\mathbf{X}$,

- What are the dimensions of each of the matrices/vectors?

- If we multiply the righthand of the expression to obtain $\mathbf{x}' = 2 \cdot \mathbf{K}\,[\mathbf{R}|\mathbf{t}]\,\mathbf{X}$, how are $\mathbf{x}$ and $\mathbf{x}'$ related?

Problem 4. What are the 3D coordinates of the image principal point expressed in the camera coordinate system?